

Question 1

Using the mtcars dataset built in R, answer the following:

- (1) Test whether the mean miles per gallon (mpg) of cars with 4 cylinders is significantly greater than 23 mpg. State your hypotheses, check the necessary assumptions, perform an appropriate one-sample test, and report your conclusion.
- (2) Use permutation sampling to simulate the same one-sample test and report the empirical p -value. (Refer to the permutation approach for a one-sample mean discussed in the lecture materials.)

Question 2

A researcher wants to determine if there is a significant improvement in student test scores after using a new study technique. The researcher recruits 15 students and measures their test scores before and after using the new study technique. Conduct a paired t-test at the 1% level of significance and plot a boxplot to visualize the distribution of the test scores before and after the treatment.

Data:

Here is the data for the test scores before and after using the new study technique:

Student	Before	After
1	75	85
2	80	82
3	70	78
4	85	88
5	65	70
6	90	95
7	78	84
8	72	80
9	88	92
10	76	81
11	84	87
12	70	73
13	68	75
14	85	90
15	77	83

Question 3

Serum creatinine(Cr) is a biomarker for assessing kidney function, and increased levels often indicate impaired renal filtration. A pharmaceutical company is conducting a clinical trial to evaluate the renal safety of a novel anti-cancer drug. Two independent groups(Treatment vs. Placebo) of patients with comparable baseline indicators (age, sex, BMI, Cr) were enrolled. After 8 weeks, the observed data for the change in serum creatinine ($\Delta Cr = Cr_{8week} - Cr_{baseline}$, unit in mg/dL) are tested and are provided below:

```
 $\Delta Cr_{placebo} = c(0.00, 0.01, -0.03, -0.01, -0.02, -0.01, 0.03, -0.02, 0.00, 0.04, 0.02, -0.02, -0.01, 0.00, -0.03)$ 
```

```
 $\Delta Cr_{treatment} = c(0.23, 0.08, 0.10, 0.15, 0.09, 0.12, 0.19, 0.08, 0.16, 0.10, 0.19, 0.12, 0.15, 0.13, 0.14)$ 
```

Does the statistical evidence suggest that this drug is associated with increased renal stress?

(hint: use "var.test" and "t.test", explicitly specify the var.equal argument in your code.)

Question4

To study the effect of two antibiotics on the diameter (mm) of bacterial inhibition zones, independent petri dishes were randomly selected for the experiment (unpaired design). It is known that the inhibition zone diameters are normally distributed, and the variances of the two groups are equal.

Experimental group (Antibiotic A): 18.5, 19.2, 17.8, 20.1, 19.5, 18.9

Control group (Antibiotic B): 15.2, 16.8, 14.9, 16.3, 15.8, 16.1

Test whether there is a significant difference in the antibacterial effect of the two antibiotics (two-tailed test, $\alpha=0.05$);

Do not use R's built-in t.test() function;

Calculate all statistics of the two-sample unpaired t-test strictly step by step.

Question 5

Researchers measured log10 viral loads in 10 cell lines before and after treatment with a novel interferon. Due to non-normality and a small sample size, use a non-parametric paired test to assess the treatment effect. The following table records the data for the 10

samples before and after treatment:

Sample ID	Before	After
1	6.5	5.8
2	7.2	7.0
3	5.8	5.6
4	8.1	7.2
5	6.9	6.1
6	7.5	7.6
7	6.3	5.5
8	7.8	7.1
9	7.0	6.8
10	6.7	6.2

1. Does this experiment follow an independent or paired sample design? Why is the Wilcoxon Signed-Rank Test chosen over the paired t-test in this scenario?
2. Using R, input the data above and perform the Wilcoxon Signed-Rank Test.
3. Create a Boxplot or Violin plot to visualize the changes in viral load before and after the treatment.
4. If the resulting p-value is less than 0.01, what does this represent in terms of biological research significance?

Hint: *In R, the `wilcox.test()` function requires the parameter `paired = TRUE` for paired data analysis.*